AGENTIC AI PROJECT

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DOMAIN: DATABASE AGENT

Problem:

Band directors have many logistical responsibilities outside of teaching and conducting music:

- Track hundreds of instruments and consumable equipment, repairs, and checkouts to students
- Selecting performance music based on student skills and instrumentation
- Financial and logistical planning for performances

Goal:

This agent will enable band directors to manage their instrument inventory, music selection, and ensemble planning, all through natural language. No SQL knowledge needed.

AGENT ARCHITECTURE & FLOW

1) Planner

- Determines the request intent and produces structured data from the request

2) Generator

- Produces SQL statement from the structured data

3) Validator

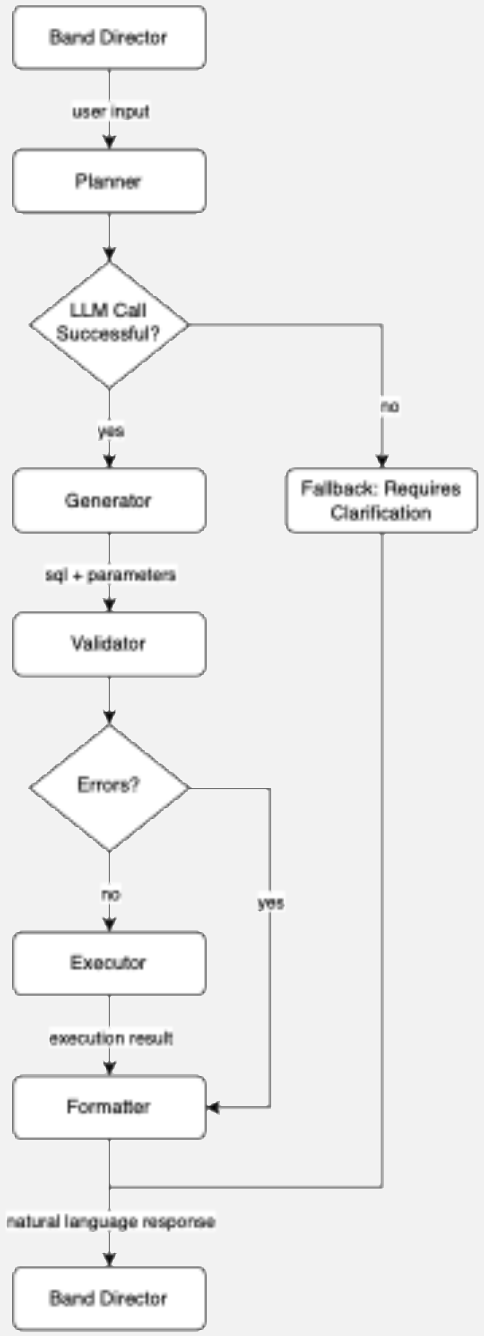
- Enforces safety and business rules before executing query

4) Executor

- Runs the validated SQL query in the database

5) Formatter

- Turns the query results into a natural language response



PROMPT TO STRUCTURED DATA

"Show me all grade 10 students who play trumpet"

```
plan = {
    "intent": "SELECT",
    "entity": "students",
    "is_batch": false,
    "records": [],
    "filters": {
        "grade": 10,
        "instrument_name": "trumpet"
    },
    "updates": {},
    "requires_clarification": false,
    "clarification_question": null
}
```

```
generated = {
    "sql": "SELECT s.student_id, s.first_name, s.last_name, s.grade FROM students s JOIN plays p ON s.s
    "params": [10, "trumpet"],
    "batch_params": [],
    "is_batch": false
}
```

SQL TO NATURAL LANGUAGE


```
{
    "success": true,
    "rows_affected": 0,
    "results": [
        {"student_id": 3, "first_name": "Emily", "last_name": "Chen", "grade": 10},
        {"student_id": 7, "first_name": "Marcus", "last_name": "Rivera", "grade": 10}
    ],
    "error": null,
    "plan": {
        "intent": "SELECT",
        "entity": "students",
        "filters": {"grade": 10, "instrument_name": "trumpet"}
    },
    "user_input": "Show me all grade 10 students who play trumpet"
}
```


```
There are 2 grade 10 students who play trumpet:
- Emily Chen
- Marcus Rivera
```

KEY TECHNICAL CHALLENGES & SOLUTIONS

Challenges

- LLM interprets the same prompts inconsistently
- LLM-based Planner can produce unexpected results with ambiguous phrasing

Solutions

- Constant incremental adjustments to the system prompts
- LLM response suggests alternate ways to phrase requests

LESSONS LEARNED & FUTURE DIRECTIONS

- Structuring the user request into a consistent format improves overall reliability
- SQL Queries should account for natural language tendencies (Woodwind vs woodwind or string vs strings)
- The Planner is the bottleneck. It doesn't always parse user prompts correctly, but when it does, everything else works as expected

Future Directions

- Continue iterating the Planner system prompts, they are still not as effective as they can be
- Implement a simple front end for real user testing
- Parse database parameters from sheet music PDFs to send to the Planner

DEMO